FSF3827 Topics in Optimization, 2025
Assignment 5
Due Wednesday January 26 2026 19.00

Instructions for the assignments

- **Collaborations.** It is fine for students to collaborate on solving the assignment. However, every student needs to submit a report and if you have collaborated with someone you need to mention this clearly in the report. If you collaborate, it is essential that you all solve each task and fully understands all parts.
- **Using AI / building upon code you found online.** It is fine to use an AI tool or build upon code that you find online. However, you need to fully understand the code and be able to discuss all parts of the code without help from AI. The teacher will ask questions and make sure you understand what the code does. If you use AI, I recommend you ask it to help with writing parts of the code or subroutines and not full programs.
- The report should have a leading title page where the name, personal number, and e-mail address is clearly stated.
- You are encourage to write the reports in Latex.
- Content
  1. The solutions must be described in enough details that other students (and the teacher) can easily read the report. For example, clearly define variables and sets used in the formulations.
  2. Describe the main parts of the problem formulations. For example, what does the constraints do.
  3. Describe the main parts of the code (what it does and how it works).
- Everyone needs to upload on the Canvas page of the course no later than by the deadline of the assignment:
  - The report as a pdf file.
  - all the relevant code in a zip file.
- You will be graded on a pass/fail scale. If the reports is insufficient, you will get second chance to revise the report.

In this final assignment you will learn about some methods for MINLP.

1. Read chapters 1 – 4 of the Convex MINLP review paper by Kronqvist, Bernal, Lundell, and Grossmann (paper available on Canvas). Read the description of the extended cutting plane (ECP) method and outer approximation (OA) more carefully. If you are interested, or need more details, the paper presenting the ECP method by Westerlund and Petterson and the OA paper by Fletcher and Leyffer are available on Canvas. Note that OA was first presented by Duran and Grossmann in 1986, but the revised version by Fletcher and Leyffer is more general.

a) Prove that the ECP method converges to an optimal solution for convex MINLP. You can assume all functions are differentiable, convex and that the linear constraints defines a bounded feasible set. You can also make other reasonable assumptions.

You can for example prove it by contradiction (assume it converges to an infeasible point) . At first, don't look at the Extended Cutting Plane paper from 1995. In my opinion, convergence can be proven a bit easier. As inspiration, read the Kelley's cutting plane paper from 1960 and see how he proved convergence (the paper is on Canvas).

b) Read *Solving mixed integer nonlinear programs by outer approximation* by Fletcher and Leyffer. Explain how they proved convergence, and the assumptions they made. It doesn't need to be a mathematically rigorous, but explain the main steps.

Consider the following small MINLP problem

$$
\begin{aligned}
\text{minimize} \quad & -\sum_{i=1}^{8} x_i \\
\text{subject to} \quad & \sum_{i=1}^{8} x_i^2 \leq 3 \\
& x_1 + x_3 + x_5 + x_7 \leq 2, \\
& -2 \leq x_i \leq 2, \quad i = 1, \ldots, 8, \\
& x_i \in \{0, 1, 2, 3\}, \quad i = 5, \ldots, 8.
\end{aligned}
$$

b) Implement the ECP algorithm and apply it to the problem. Stop the algorithm when the MILP relaxation gives a solution that violates the nonlinear constraint by less than 0.0001. Solve all the subproblem by a suitable solver, for example, Gurobi. How many iterations and time does it take to solve the problem?

The algorithm will need approximately 100 iterations, but it is worth mentioning that the algorithm can be modified to be much more efficient. For example, one can add the cuts through cut callbacks (similar as we did for TSP). That way we only solve one MILP subproblem where cuts are added as needed. For the algorithm to be practical, one also needs some technique for computing feasible solutions (ECP gives valid lower bounds in each iteration, but it doesn't give feasible solutions). But, you don't need to implement such techniques.

c) Implement the OA algorithm and apply it to the problem. You can solve all the subproblems using Gurobi. (You could solve the problem directly with Gurobi. But, the point of this exercise is for you to learn by implementing the algorithm)

2. Now consider the problem

$$
\begin{aligned}
\text{minimize} \quad & -\sum_{i=1}^{8} x_i \\
\text{subject to} \quad & \sum_{i=1}^{8} z_i \leq 3 \\
& x_i^2 \leq z_i, \quad i = 1, \dots, 8, \\
& x_1 + x_3 + x_5 + x_7 \leq 2, \\
& -2 \leq x_i \leq 2, \quad i = 1, \dots, 8, \\
& 0 \leq z_i, \quad i = 1, \dots, 8, \\
& x_i \in \{0, 1, 2, 3\}, \quad i = 5, \dots, 8,
\end{aligned}
$$

which can be considered a lifted reformulation of the problem in Question 1. Both problems have the same optimal x-variables. The main difference is that we have now separated the convex nonlinear constraint into simpler convex constraints with fewer variables. As mentioned during a lecture, this can result in much stronger outer approximations of the feasible set.

- Solve this lifted problem using ECP. You can choose to generate cuts for all nonlinear constraints in each iteration or only cuts for the violated constraints.
- Solve this lifted problem using OA.

Does this formulation of the problem solve more efficiently? Advanced solvers automatically applies this type of reformulation and many others to solve problems more efficiently.

3. Read the *Extra material on global optimization* (attached at the end).

Consider the following nonconvex problem

$$
\begin{aligned}
\text{minimize} \quad & (x_1 + x_2)^2 - x_2 x_3 - 0.25\, x_4^2 + x_4 x_5 \\
\text{subject to} \quad & x_1 + x_2 + x_3 + x_4 + x_5 \leq 7.5, \\
& x_1 + x_5 \geq 2, \\
& x_1 + x_4 \geq 1, \\
& x_2 + 0.25\, x_3 \leq 2.5, \\
& 0 \leq x_i \leq 3, \quad i = 1, \dots, 5.
\end{aligned}
$$

a) Apply the factorable programming approach to construct a convex relaxation of the problem. Note that you don't need to relax convex nonlinear terms (there is something convex in the objective). Solve the convex relaxation. The optimal objective value for the problem should be -0.5, but you will most likely get a smaller objective value from the relaxation.

b) Read chapter $1-2$ in the paper *A New Reformulation-Linearization Technique for Bilinear Programming Problems*. Explain how the RLT concept could be used to construct a convex relaxation for this problem. **You don't need to implement it**. If you want an additional challenge, implement it and try to construct a stronger relaxation.

Good luck!

Extra material on global optimization

January 2026

This material is intended as supplemental material and some repetition of lecture 9. As some people missed that lecture, I want to give some material to read on this. **Disclaimer:** The teacher has used ChatGPT to generate parts of this text.

Introduction

This document briefly covers some fundamental concepts from global optimization. Global optimization is a subfield of mathematical programming that focuses on finding globally optimal solutions to nonconvex problems.

Convex and concave envelopes.

Let $f : D \to R$ be a real-valued function defined on a set $D \subset R^n$.

The *convex envelope* of f on D, denoted by $\mathrm{vex}_D\, f$, is defined as

$$\mathrm{vex}_D\, f(x) = \sup \left\{ g(x) : \begin{array}{l} g \text{ is convex on } D, \\ g(y) \leq f(y) \text{ for all } y \in D \end{array} \right\}.$$

Intuitively, the convex envelope is the *largest convex function that underestimates f on D.* Among all convex underestimators, it lies as close as possible to f everywhere. In minimization problems, replacing f by its convex envelope yields the tightest possible convex relaxation over D (assuming there are no other nonconvex parts).

Similarly, the *concave envelope* of f on D, denoted by $\mathrm{cav}_D\, f$, is defined as

$$\mathrm{cav}_D\, f(x) = \inf \left\{ h(x) : \begin{array}{l} h \text{ is concave on } D, \\ h(y) \geq f(y) \text{ for all } y \in D \end{array} \right\}.$$

It is the *smallest concave function that overestimates f on D.*

Convex and concave envelopes play a central role in global optimization, since they are *optimal relaxations*: no other convex underestimator or concave overestimator can be tighter for the function f over D.

McCormick Envelopes

McCormick envelopes are named after Garth P. McCormick, and can be derived from the framework that he presented in the paper ”Computability of Global Solutions To Factorable Nonconvex Solutions: Part I: Convex Underestimating Problems ” in Mathematical Programming 1976.

The McCormic envelopes are used in global optimization for constructing convex relaxations of bilinear terms. They provide an exact polyhedral description of the convex hull of a bilinear function over a rectangular domain and are widely used in deterministic global optimization and mixed-integer nonlinear programming.

Consider the bilinear function

$$f(x, y) = xy$$

defined on the box

$$\mathcal{B} = [x_L, x_U] \times [y_L, y_U],$$

where $x_L < x_U$ and $y_L < y_U$. The McCormic envelopes are 4 linear inequalities that describes bot the convex envelope and concave envelope of f over $\mathcal{B}$. Two of the inequalities forms the convex envelope and the other two forms the concave envelope.

A convex underestimator of f on $\mathcal{B}$ is a convex function g satisfying

$$g(x, y) \leq xy \quad \text{for all } (x, y) \in \mathcal{B},$$

and the maximal among all such functions is the convex envelope. Analogously, a concave overestimator is a concave function h such that

$$h(x, y) \geq xy \quad \text{for all } (x, y) \in \mathcal{B}.$$

The McCormick inequalities can be derived directly from the variable bounds. By combining two of the variable bounds we can construct the valid inequality

$$(x - x_L)(y - y_L) \geq 0.$$

Note that the inequality is always true for all $(x, y) \in \mathcal{B}$. Expanding this inequality yields

$$xy \geq x_L y + x y_L - x_L y_L.$$

Note that right hand side gives us an underestimator of xy.

Similarly, from

$$(x_U - x)(y_U - y) \geq 0,$$

we obtain

$$xy \geq x_U y + x y_U - x_U y_U.$$

Two additional inequalities are obtained by considering the remaining sign-consistent products of bounds:

$$(x_U - x)(y - y_L) \geq 0, \qquad (x - x_L)(y_U - y) \geq 0,$$

which expand to

$$xy \leq x_U y + x y_L - x_U y_L,$$
$$xy \leq x_L y + x y_U - x_L y_U.$$

Introducing an auxiliary variable z to represent the bilinear term, the four McCormick envelope inequalities can be written as

$$\begin{aligned} z &\geq x_L y + x y_L - x_L y_L, \\ z &\geq x_U y + x y_U - x_U y_U, \\ z &\leq x_U y + x y_L - x_U y_L, \\ z &\leq x_L y + x y_U - x_L y_U. \end{aligned}$$

Geometrically, each inequality corresponds to a supporting hyperplane of the surface $z = xy$ at one of the four corners of the box $\mathcal{B}$. The intersection of these four half-spaces is exactly the convex hull of the bilinear graph over $\mathcal{B}$. Consequently, the McCormick envelopes are the tightest possible linear relaxation of a single bilinear term using only bound information.

As variable bounds are tightened, for example within a branch-and-bound framework, the envelopes gets tighter and tighter. This property makes McCormick envelopes a cornerstone of deterministic global optimization methods.

Factorable Programming

Many problems have more complex terms than bilinear functions, and convex/concave envelopes are generally only known for simple functions (e.g., functions of one, or two variables). This is where factorable programming comes in. Factorable programming addresses nonconvex optimization problems by exploiting the fact that many nonlinear functions can be written as compositions of simple elementary operations such as addition, multiplication, and univariate nonlinear functions. A function is called factorable if it admits such a recursive decomposition.

This structure allows nonconvex problems to be reformulated using auxiliary variables to break up the nonlinear functions into simpler parts with known convex and concave envelopes (or convex/concave under- and over estimators). This enables the nonconvex terms to be relaxed individually.

Illustrative example

Consider the problem

$$\begin{aligned} \text{minimize} \quad & (x_1 x_2 + 1)\, x_3 - x_4^2 \\ \text{subject to} \quad & 0 \leq x_1 \leq 1, \\ & 0 \leq x_2 \leq 1, \\ & 0 \leq x_3 \leq 1, \\ & L \leq x_4 \leq U. \end{aligned}$$

First, we introduce auxiliary variables

$$w = x_1 x_2, \qquad z = w + 1, \qquad y = z x_3.$$

This allows us to write the problem as

$$\begin{aligned} \text{minimize} \quad & y - x_4^2 \\ \text{subject to} \quad & w = x_1 x_2, \\ & z = w + 1, \\ & y = z x_3, \\ & 0 \le x_1 \le 1, \\ & 0 \le x_2 \le 1, \\ & 0 \le x_3 \le 1, \\ & L \le x_4 \le U. \end{aligned}$$

We have now separated the nonconvex part into terms that are simple enough for us.

We start by using the McCormic envelopes to relax the equations with bilinear term. We first focus on $w = x_1 x_2$. Since $x_1, x_2 \in [0, 1]$, the McCormick envelopes for $w = x_1 x_2$ are given by

$$\begin{aligned} & w \ge 0, \\ & w \ge x_1 + x_2 - 1, \\ & w \le x_1, \\ & w \le x_2. \end{aligned}$$

We continue with $y = z x_3$, but then we also need bounds for z. From the constraints we know $w \in [0, 1]$, and since

$$z = w + 1$$

we get

$$z \in [1, 2].$$

Next, consider the bilinear relation $y = z x_3$ with $z \in [1, 2]$ and $x_3 \in [0, 1]$. The corresponding McCormick envelopes are

$$\begin{aligned} & y \ge x_3, \\ & y \ge 2x_3 + z - 2, \\ & y \le 2x_3, \\ & y \le x_3 + z - 1. \end{aligned}$$

Next, we focus on the objective term $-x_4^2$. On the interval $[L, U]$, the function $-x_4^2$ is concave and lies above its secant. The secant line is actually the convex envelope of a univariate concave function over the interval $[L, U]$. The secant line passing through the points $(L, -L^2)$ and $(U, -U^2)$ is

$$\ell(x_4) = -(L + U)x_4 + LU.$$

This affine function satisfies

$$\ell(x_4) \le -x_4^2 \qquad \text{for all } x_4 \in [L, U].$$

Therefore, a valid convex relaxation for a minimization problem is obtained by replacing the term $-x_4^2$ with $\ell(x_4)$.

Note that, since $-x_4^2$ is in the objective and we are minimizing we only need to relax the function in one direction (we will always try to make $-x_4^2$ as small as possible). If $-x_4^2$ was in an equality constraint we would need both the convex envelope and concave envelope.

We then get the convex relaxation

$$\begin{aligned}
\text{minimize} \quad & y \;-\; (L + U)\, x_4 \;+\; LU \\
\text{subject to} \quad & w \ge 0, \\
& w \ge x_1 + x_2 - 1, \\
& w \le x_1, \\
& w \le x_2, \\
& z = w + 1, \\
& y \ge x_3, \\
& y \ge 2x_3 + z - 2, \\
& y \le 2x_3, \\
& y \le x_3 + z - 1, \\
& 0 \le x_1 \le 1, \\
& 0 \le x_2 \le 1, \\
& 0 \le x_3 \le 1, \\
& L \le x_4 \le U, \\
& 0 \le w \le 1, \\
& 1 \le z \le 2, \\
& 0 \le y \le 2.
\end{aligned}$$